

Operating Systems

BCA — Semester IV — 2021 — Answer Sheet
Subject Code: CACS251

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group C Attempt any TWO questions[2x10=20]

2. What is an Operating System? Why is it known as the resource manager and the extended machine?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is Kernel? What role does it play in an Operating System? Differentiate between monolithic and micro kernel.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Suppose that a disk has 200 cylinders (0-200). The head pointer is currently at 50 and previously at 35. The queue of pending request, in FIFO order is: 60, 190, 40, 125, 15, and 150. The time required to move per cylinder is 6 msec. Starting from the current head position, what is the total seek time needed to satisfy all the pending request for each of the following disk scheduling algorithms? a) FCFS b) SCAN

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define file and directory. Explain the concept of Access Control List (ACL) and Access Control Matrix (ACM).

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is biometric password in authentication? Explain the various system threats.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. How distributed operating system is more applicable than centralized operating system? Explain the major goals of distributed operating system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Write short notes on any TWO : a) Producer Consumer problem b) Coalescing and Compaction c) Ubuntu

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is CPU scheduling? Write down the criteria for CPU scheduling? Consider the following set of processes, with the length of the CPU burst given in milliseconds, draw Gantt chart illustrating their execution and calculate average waiting time and turnaround time using:a) First Come First Serveb) Shortest Job Firstc) Non-preemptive priority (smaller number implies higher priority)d) Round Robin (quantum = 1)

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define the terms page fault and thrashing. Consider the following page reference string: 1, 3, 5, 1, 7, 1, 5, 5, 1, 4, 2, 7, 6, 3, 4, 1. How many page faults would occur for each of the following page replacement algorithms assuming 3 page frames?a) FIFO page replacementb) LRU page replacementc) Optimal page replacement

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Define deadlock. List out the conditions that can result in resource deadlock. Explain different deadlock handling methods in detail.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.